Grade 10 Practice

Standard G.CO.1

Computation

1. How many dimensions does a point have? ___ **0**

2. How many directions does a line extend? ___ **2**

3. An angle bisector divides an angle into how many equal angles? ___ **2**

4. How many endpoints does a segment have? ___ **2**

Word problems

5. How many endpoints does a ray have?
Answer: 1

6. Two lines in the same plane that never meet are called what?
Answer: parallel

Grade 10 Practice

Standard G.CO.6

Computation

1. Does a reflection preserve distance? Write yes or no. **yes**

2. Does a dilation preserve angle measures? Write yes or no. **yes**

3. Does a dilation preserve distance? Write yes or no. **no**

4. Is a figure congruent to its rotation? Write yes or no. **yes**

Word problems

5. A triangle is reflected across a line. Is the image congruent to the original? Write yes or no.
Answer: yes

6. A triangle is dilated by a factor of 2. Is the image congruent to the original? Write yes or no.
Answer: no

Grade 10 Practice

Standard G.CO.7

Computation

1. If all three pairs of corresponding sides are equal, are the triangles congruent? Write

2. If two sides and the included angle are equal, are the triangles congruent by SAS? W

3. If only two angles are equal, are the triangles congruent? Write yes or no. **no**

4. Is SSA always enough to prove congruence? Write yes or no. **no**

Word problems

5. Triangles have sides 3, 4, and 5 and another has sides 3, 4, and 5. Are they congruent? Write yes or no.
Answer: yes

6. Two triangles have two equal angles and one equal side. Are they congruent by AAS? Write yes or no.
Answer: yes

Grade 10 Practice

Standard G.CO.8

Computation

1. Two triangles have all three sides equal. Name the criterion: ___ **SSS**

2. Two triangles have two sides and the included angle equal. Name the criterion: ___

3. Two triangles have two angles and the included side equal. Name the criterion: ___

4. Two right triangles have equal hypotenuses and one equal leg. Name the criterion: _

Word problems

5. Two triangles share a side and have two pairs of equal angles. Which criterion applies, ASA or AAS?
Answer: ASA or AAS

6. Two right triangles have equal hypotenuses and one equal leg. Name the criterion.
Answer: HL

Grade 10 Practice

Standard G.CO.9

Computation

1. Vertical angles are equal. Write yes or no. **yes**

2. Angles on a straight line add up to how many degrees? ___ **180**

3. When two parallel lines are cut by a transversal, alternate interior angles are equal. V

4. The angles of a triangle add up to how many degrees? ___ **180**

Word problems

5. Angles around a point add up to how many degrees?
Answer: 360 degrees

6. Corresponding angles are equal when the lines are what?
Answer: parallel

Grade 10 Practice

Standard G.CO.10

Computation

1. In an isosceles triangle, the base angles are ___ equal. **equal**

2. A midsegment of a triangle is ___ the length of the third side. **half**

3. A triangle has angles 50° and 60°. What is the third angle? ___ **70°**

4. An exterior angle equals the sum of two remote interior angles of 60° and 50°. What

Word problems

5. In an isosceles triangle, one base angle is 40°. What is the vertex angle?
Answer: 100°

6. A triangle has a base of 10. What is the length of its midsegment parallel to the base?
Answer: 5

Grade 10 Practice

Standard G.CO.11

Computation

1. Opposite sides of a parallelogram are equal. Write yes or no. **yes**

2. Diagonals of a rectangle are equal. Write yes or no. **yes**

3. Diagonals of a rhombus are perpendicular. Write yes or no. **yes**

4. Consecutive angles of a parallelogram add up to how many degrees? ___ **180**

Word problems

5. A parallelogram has an angle of 70°. What is an adjacent angle?
Answer: 110°

6. In a rectangle, the diagonals bisect each other. Do their midpoints match? Write yes or no.
Answer: yes

Grade 10 Practice

Standard G.SRT.2

Computation

1. Two triangles have two pairs of equal angles. Are they similar? Write yes or no. **yes**

2. A triangle has a scale factor of 3 from another. What is the ratio of their perimeters?

3. Triangles with angles 40° and 60° and 40° and 80° — are they similar? Write yes or n

4. Triangles with side ratios 1:2 and 2:4 — are they similar? Write yes or no. **yes**

Word problems

5. Triangles have sides 6, 8, and 10, and others have sides 3, 4, and 5. Are they similar? Write yes or no.
Answer: yes

6. A triangle has a side of 5 and is dilated by a scale factor of 2. What is the new side length?
Answer: 10

Grade 10 Practice

Standard G.SRT.4

Computation

1. Triangle with sides 3, 4, and 5. Is it a right triangle? Write yes or no. **yes**

2. Triangle with sides 5, 12, and 13. Is it a right triangle? Write yes or no. **yes**

3. Triangle with sides 4, 5, and 6. Is it a right triangle? Write yes or no. **no**

4. Two triangles have two equal angles. What theorem makes them similar? Write AA.

Word problems

5. A ladder's shadow and a tree's shadow form similar triangles. What theorem proves it when two angles match?
Answer: AA

6. Two right triangles share one acute angle. Are they similar? Write yes or no.
Answer: yes

Grade 10 Practice

Standard G.SRT.5

Computation

1. Similar triangles have sides 3 and 6. A side of 4 in the smaller triangle matches what

2. Ratio of similar triangles is 2:5. A side of 10 in the smaller matches what side in the la

3. A 2 m post casts a 1 m shadow when a 10 m tree casts a 5 m shadow. How does this

4. Scale factor 1.5 applied to side 8. What is the new side? ___ **12**

Word problems

5. A 6 ft person casts a 4 ft shadow at the same time a tree casts a 20 ft shadow. How tall is the tree?
Answer: 30 feet

6. Two similar triangles have sides 4 and 6. If the smaller triangle has a side of 10 that matches the 4, what is the matching side in the larger?
Answer: 15

Grade 10 Practice

Standard G.SRT.6

Computation

1. In a right triangle, sin = opposite / ___ **hypotenuse**

2. Find cos 60°. ___ **1/2**

3. Find tan 45°. ___ **1**

4. Find sin 90°. ___ **1**

Word problems

5. A right triangle has opposite side 3 and adjacent side 4. What is tan A?
Answer: 3/4

6. A right triangle has hypotenuse 10 and opposite side 6. What is sin A?
Answer: 3/5

Grade 10 Practice

Standard G.SRT.8

Computation

1. A right triangle has legs 3 and 4. Find the hypotenuse. ___ **5**

2. A right triangle has hypotenuse 10 and angle 30°. Find the opposite side. ___ **5**

3. A right triangle has legs 5 and 12. Find the hypotenuse. ___ **13**

4. A right triangle has adjacent side 6 and angle 45°. Find the opposite side. ___ **6**

Word problems

5. A ramp is 10 m long at an angle of 15°. About how high is it? Round to the nearest hundredth.
Answer: 2.59 m

6. A ladder 13 ft long reaches a wall 12 ft high. How far is its base from the wall?
Answer: 5 ft

Grade 10 Practice

Standard G.C.2

Computation

1. A central angle measures 80°. What is the measure of its arc? ___ **80°**

2. An inscribed angle is ___ its intercepted arc. Write half or double. **half**

3. An inscribed angle intercepts a 100° arc. What is the inscribed angle? ___ **50°**

4. An inscribed angle intercepts a semicircle. What is its measure? ___ **90°**

Word problems

5. A central angle is 120°. What is an inscribed angle on the same arc?
Answer: 60°

6. A central angle is 70°. What is an inscribed angle on the same arc?
Answer: 35°

Grade 10 Practice

Standard G.C.5

Computation

1. A circle has radius 6 and a 60° central angle. What is the arc length? Write in terms o

2. A sector has radius 6 and a 90° angle. What is its area? Write in terms of π. **9π**

3. A circle has radius 3. What is its circumference? Write in terms of π. **6π**

4. A circle has radius 10 and a 36° arc. What is the arc length? Write in terms of π. **2π**

Word problems

5. A pizza has radius 8 in. A 1/4 slice is cut. What is the arc length of the crust? Write in terms of π.

Answer: 4π inches

6. A sector has radius 4 and a 180° angle. What is its area? Write in terms of π.

Answer: 8π

Grade 10 Practice

Standard G.GPE.4

Computation

1. What is the midpoint of (0, 0) and (4, 6)? ___ **(2, 3)**

2. What is the distance from (0, 0) to (3, 4)? ___ **5**

3. What is the slope of the line from (0, 0) to (3, 4)? ___ **4/3**

4. Is the point (1, 2) on the line y = 2x? Write yes or no. **yes**

Word problems

5. Parallelogram ABCD has segments AB and CD with equal slopes. Are they parallel? Write yes or no.
Answer: parallel

6. What is the midpoint of (1, 1) and (3, 5)?
Answer: (2, 3)

Grade 10 Practice

Standard G.GPE.6

Computation

1. What point divides (0, 0) to (4, 8) into a 1:1 ratio? ___ **(2, 4)**

2. What point divides (0, 0) to (6, 0) into a 1:2 ratio? ___ **(2, 0)**

3. What is the midpoint of (-2, 4) and (2, 8)? ___ **(0, 6)**

4. What point is 1/4 of the way from (0, 0) to (8, 0)? ___ **(2, 0)**

Word problems

5. What is the midpoint of the segment from (1, 1) to (5, 9)?
Answer: (3, 5)

6. What point is 2/3 of the way from (0, 0) to (9, 0)?
Answer: (6, 0)

Grade 10 Practice

Standard G.GPE.7

Computation

1. A rectangle has corners (0, 0), (5, 0), (5, 3), and (0, 3). What is its area? ___ **15**

2. A right triangle has vertices (0, 0), (4, 0), and (0, 3). What is its perimeter? ___ **12**

3. What is the distance from (0, 0) to (5, 12)? ___ **13**

4. A square has side 6 units. What is its perimeter? ___ **24**

Word problems

5. A rectangle has corners (1, 2), (7, 2), (7, 6), and (1, 6). What is its perimeter?
Answer: 20 units

6. A triangle has vertices (0, 0), (6, 0), and (3, 4). What is its area?
Answer: 12 square units

Grade 10 Practice

Standard G.GMD.1

Computation

1. What is the circumference of a circle with radius 5? Write in terms of π. **10π**

2. What is the area of a circle with radius 4? Write in terms of π. **16π**

3. What is the area of a circle with diameter 6? Write in terms of π. **9π**

4. What is the circumference of a circle with diameter 8? Write in terms of π. **8π**

Word problems

5. A tree trunk has a diameter of 10 feet. What is its circumference? Write in terms of π.
Answer: 10π feet

6. A circular rug has radius 3 feet. What is its area? Write in terms of π.
Answer: 9π square feet

Grade 10 Practice

Standard G.GMD.3

Computation

1. What is the volume of a cylinder with radius 2 and height 6? Write in terms of π. **24π**

2. What is the volume of a cone with radius 3 and height 6? Write in terms of π. **18π**

3. What is the volume of a sphere with radius 3? Write in terms of π. **36π**

4. What is the volume of a cylinder with radius 1 and height 10? Write in terms of π. **10π**

Word problems

5. A can has radius 4 cm and height 10 cm. What is its volume? Write in terms of π.
Answer: 160π cubic cm

6. A cone has radius 3 cm and height 4 cm. What is its volume? Write in terms of π.
Answer: 12π cubic cm

Grade 10 Practice

Standard G.MG.1

Computation

1. A can is modeled as a cylinder. What shape is the model? ___ **cylinder**

2. A tent is modeled as a triangular prism. Name the three dimensions needed to find i

3. A box is modeled as a rectangular prism 2 by 3 by 4. What is its volume? **24**

4. A cone is modeled with radius 2 and height 9. What is its volume? Write in terms of π

Word problems

5. A swimming pool is modeled as a rectangular prism 20 ft by 10 ft by 4 ft. What is its volume in cubic feet?
Answer: 800 cubic feet

6. A grain silo is modeled as a cylinder with radius 5 and height 20. What is its volume? Write in terms of π.
Answer: 500π cubic units